

01st-10-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

Prepared by: Aminu Ibrahim Jalo
Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Title: INTRODUCTION TO MODBUS

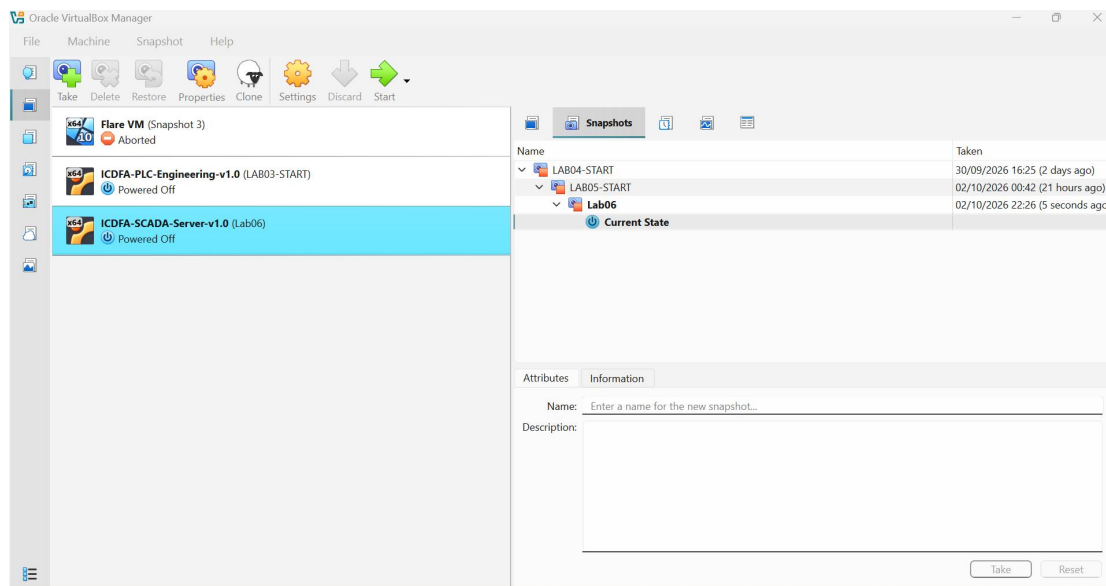
Assignment: Week 1-Lab 2 Report
C11/26/ACIS/17359

Instructor : Aminu Idris

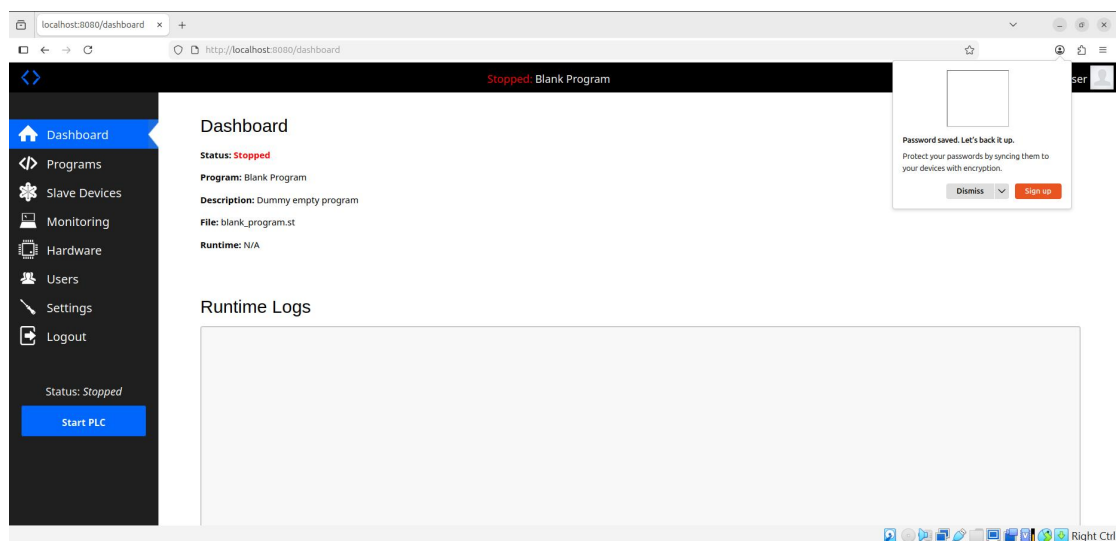
Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	01 st -10-2026

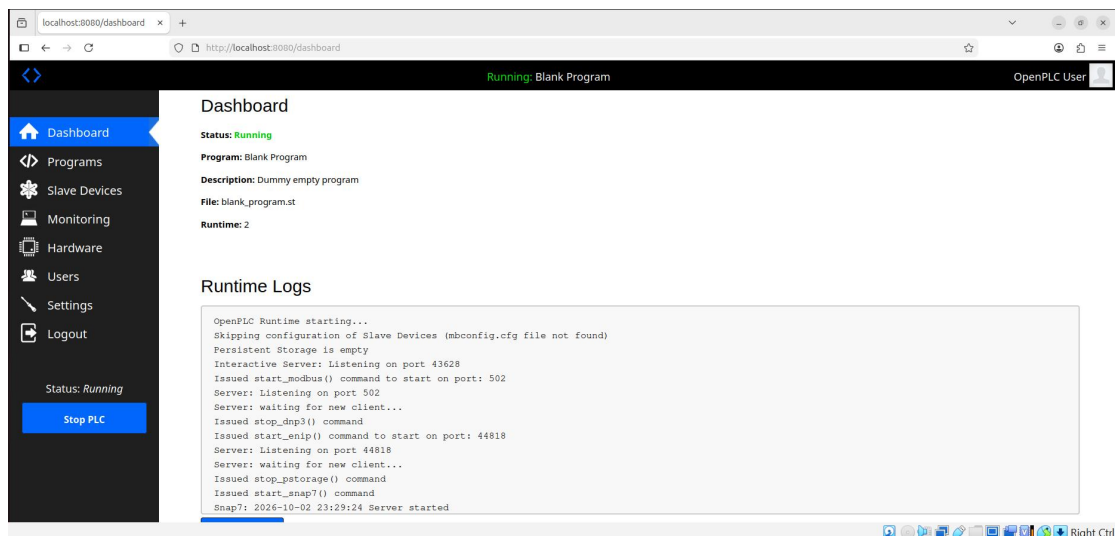
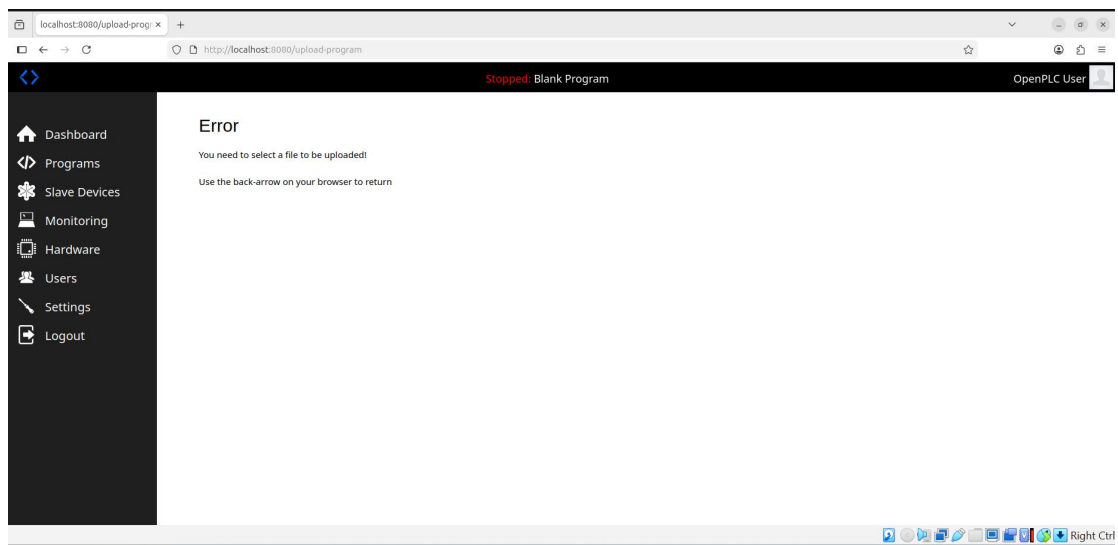
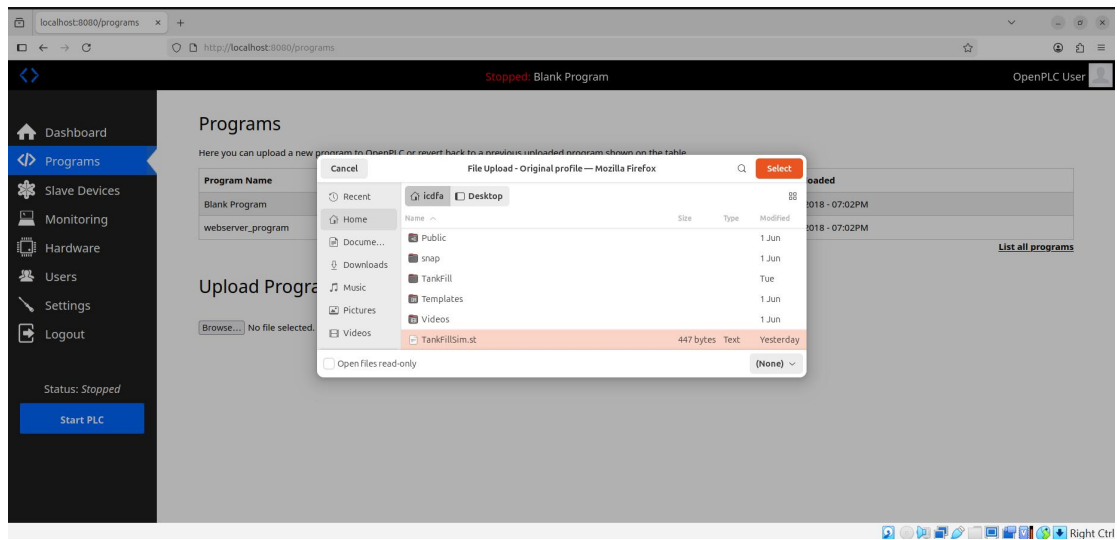
Procedure

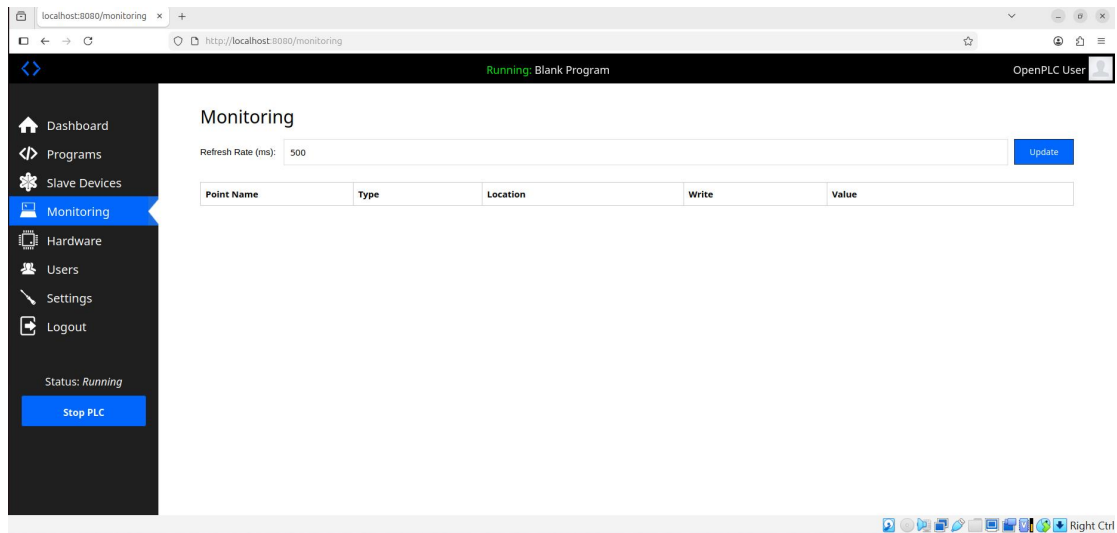
Step 1. Start the PLC, SCADA and analyst VMs on the instructor-provided isolated laboratory topology. Do not connect this lab to production or bridged networks.



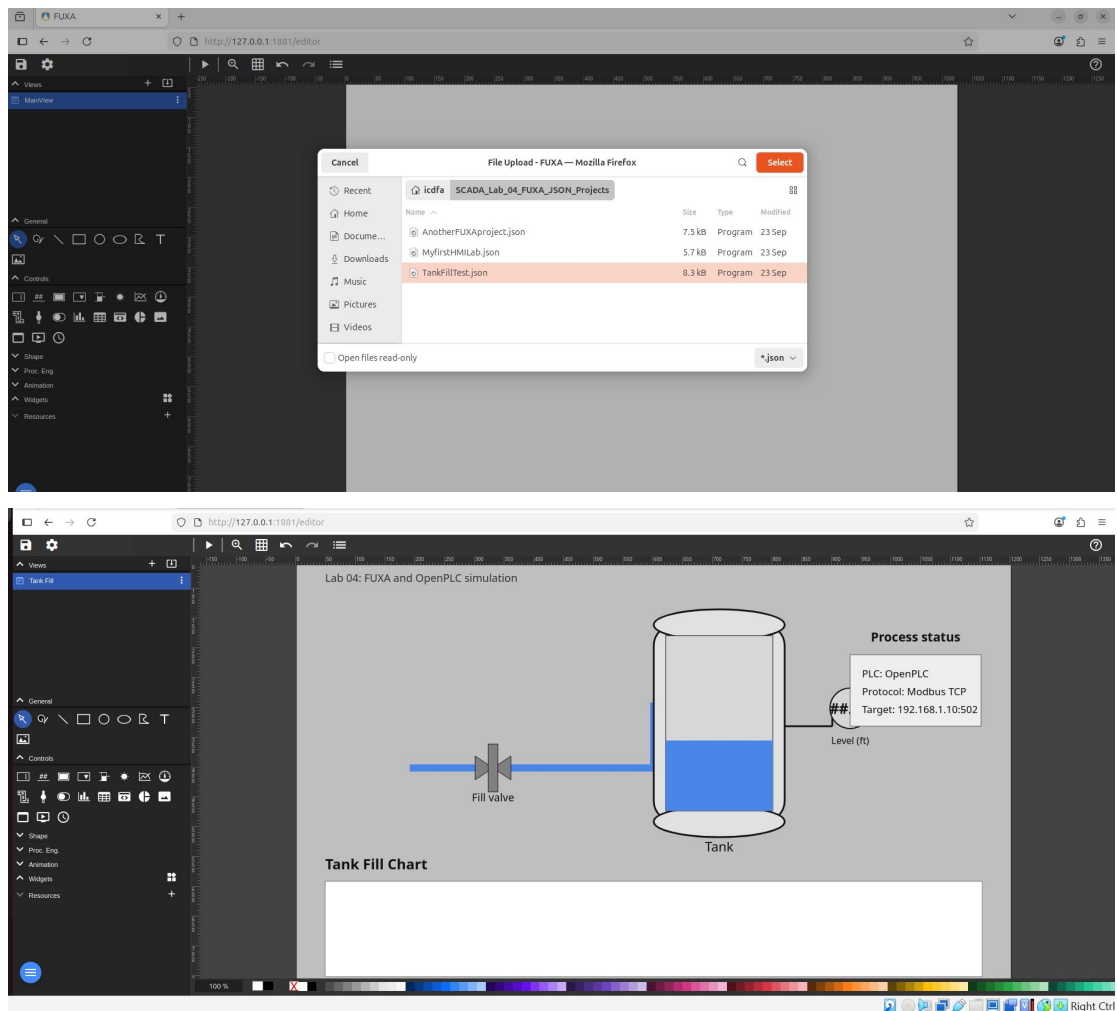
Step 2. On the PLC VM, start the TankFillSim program in OpenPLC Runtime and confirm Running status.

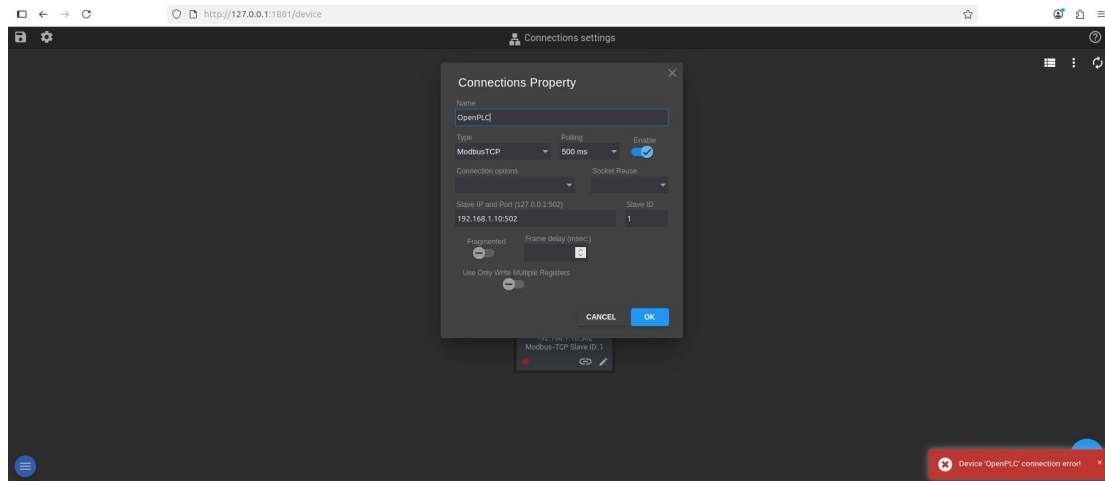




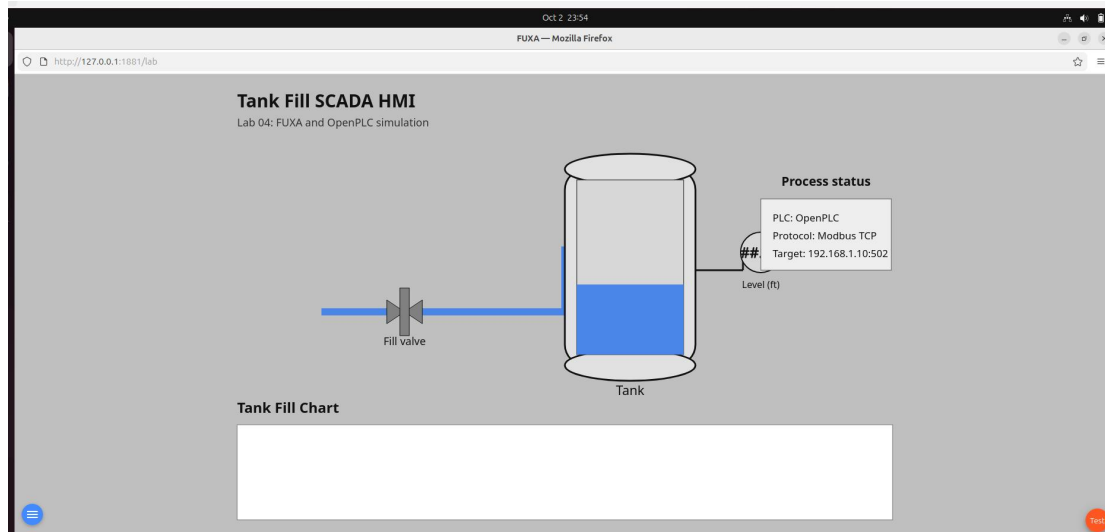


Step 3. On the SCADA VM, start FUXA, load the tank-fill HMI and confirm the openplc connection targets 192.168.1.10:502.





Step 4. Run the HMI and verify that the tank level changes between its operating limits.



Step 5. On the analyst VM, open Wireshark. Select only the interface connected to the ICDFA lab segment carrying PLC/SCADA traffic.

Step 6. Apply the display filter modbus. If needed, refine it with tcp.port == 502.

Step 7. Start a short packet capture while the HMI is running, allow several process updates to occur, then stop the capture.

Step 8. Select a Modbus request/response pair and expand the Modbus/TCP and Modbus protocol fields.

Step 9. Identify the transaction/request, source and destination IP addresses, function code and referenced register/data value.

Step 10. Locate a Function Code 4 (Read Input Registers) response associated with the tank-level value. In the teaching image, Input Register 80 corresponds to the Tank Level indication.

Step 11. Complete the Modbus function-code table by explaining at least codes 01, 02, 03, 04, 05, 06, 15 and 16.

Step 12. Save the packet capture with a filename containing your registration number and LAB06. Do not capture unrelated traffic.

Knowledge Check

1. What transport/networking difference separates Modbus TCP from Modbus RTU?
Response: Modbus TCP uses TCP/IP networks, normally over Ethernet and TCP port 502, while Modbus RTU is a serial communication protocol commonly transmitted over RS-485 or RS-232. Modbus TCP uses an Ethernet/TCP frame structure, whereas Modbus RTU uses serial frames with an RTU-specific format and CRC error checking.
2. Why does Modbus use function codes?
Response: Function codes identify the specific operation requested by the Modbus client or performed by the server. They indicate whether the communication is reading coils, reading registers, writing a coil, writing a register, or performing another supported operation.
3. What does Function Code 4 request/read?
Response: Function Code 04, or Read Input Registers, requests the values of one or more 16-bit input registers from a Modbus device. In this laboratory, Function Code 04 is used to read the tank-level indication associated with Input Register 80.
4. Why is visibility of unprotected industrial protocol traffic a cybersecurity concern?
Response: Unprotected Modbus traffic can expose device addresses, function codes, register addresses and process values to anyone with sufficient network visibility. Because Modbus does not provide built-in encryption or authentication, an unauthorized party who gains network access may be able to observe sensitive industrial communications and, depending on network controls, potentially attempt unauthorized operations. Network segmentation, access controls, monitoring and other compensating controls can reduce this risk.